

Exact Tree–Theta–Trinet Collisions under the Kimura 2- and 3-Parameter Models

Alec Kriebel

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Abstract

We construct exact three-leaf distributions shared by a tree and a binary semi-directed strict level-two theta trinet under both the Kimura two-parameter (K2P) and three-parameter (K3P) substitution models. The K2P example is a compact factorization over $\mathbb{Q}(\sqrt{71})$ with every transition probability strictly positive. Under the conventions stated in Brits, Holtgreffe, van Iersel, and Martin, it conflicts with the conclusion of their Lemma 5.6 and with the K2P case of Corollary 5.8. A fixed-order audit identifies an incompatible relabelling step in the proposed K2P induction. A second exact K2P witness lies in the strict continuous-time rate cone and is independently verified by ordinary-state Markov pruning. The fixed K2P theta map has full ambient rank 9 at both collisions; near the simple witness, its network-parameter collision locus with the 6-dimensional tree model has dimension 17 and codimension 3. An eight-parameter symmetric ansatz contains an exact six-dimensional local collision family. The K3P construction answers the source paper’s high-level K3P trinet question negatively. Its fixed theta map has full rank 15, its local collision locus has dimension 23, and a real-analytic implicit-function argument moves the collision into the strict continuous-time K3P cone. The present examples do not test the source paper’s JC or level-one results. Standard-library verifiers reconstruct every exact identity, inequality, determinant, and direct-pruning comparison.

1 Introduction

Identifiability asks whether an exact leaf-pattern distribution determines the underlying phylogenetic topology. Brits, Holtgreffe, van Iersel, and Martin established full level-one identifiability results under JC, K2P, and K3P, proved arbitrary-level tree–trinet distinguishability under JC and K2P, and used the latter to state a global tree–network distinguishability corollary under those two models [1]. Their Discussion asks whether the high-level trinet inequality and quartet distinguishability extend to K3P, observing that a positive extension appears plausible after the level-one K3P calculation.

We report two exact collision phenomena that refine this picture. First, the arbitrary-level K2P statement does not hold on a strict level-two theta trinet. The compact witness in Section 3 lies in the source paper’s stated stochastic space and is exactly a tree distribution. A separate sign point and a displayed-child audit explain why the order-sensitive level-one invariant cannot be used recursively in one fixed leaf order. Second, the same theta topology has an exact K3P tree collision, answering the high-level K3P question negatively.

The collisions are not isolated numerical curiosities. Under K2P, the fixed theta map has full rank in the nine-dimensional affine K2P Fourier space, while the tree model has dimension six. Under K3P the corresponding dimensions are fifteen and nine, and the theta map again has full ambient rank. The resulting collision loci are positive-dimensional but proper in network parameter

space. These calculations imply neither universal distinguishability nor generic indistinguishability: on this theta topology, tree equivalence is nongeneric but occurs along substantial exact families.

The present examples concern the arbitrary-level K2P trinet/global statements and the open high-level K3P trinet question. They do not test the JC trinet inequality, the level-one K2P sunlet calculation, the level-one full-identifiability theorem, the combinatorial restriction lemma, K3P quartet distinguishability, or every possible restricted level-two theorem.

Main results.

- An exact K2P tree-theta collision over $\mathbb{Q}(\sqrt{71})$, with all transition probabilities positive and exact minimum pattern probability $1188799/79626240$.
- A fixed-order diagnosis and an independent rational theta point for which the proposed K2P invariant is negative in all six leaf orders.
- An exact strict continuous-time K2P collision, verified by both Fourier inversion and direct Markov pruning.
- Full K2P rank 9 at both exact collisions, a local 17-dimensional collision locus, and an exact six-dimensional symmetric collision family.
- An exact K3P collision over $\mathbb{Q}(5^{-1/4})$, full rank 15, a local 23-dimensional collision locus, and a strict continuous-time analytic extension.

2 Models and the theta topology

Let $G = \{A, C, G, T\} \cong \mathbb{Z}_2^2$, with A the identity and $C + G = T$. A three-leaf Fourier coordinate q_{xyz} can be nonzero only when $x + y + z = A$. Thus sixteen of the sixty-four Fourier coordinates are consistent. The finite character transform is invertible, so equality of all Fourier coordinates is equivalent to equality of all sixty-four ordinary site-pattern probabilities.

A K3P edge has Fourier vector $a = (1, a_C, a_G, a_T)$. Its transition probabilities are

$$\begin{aligned} p_A &= (1 + a_C + a_G + a_T)/4, & p_C &= (1 + a_C - a_G - a_T)/4, \\ p_G &= (1 - a_C + a_G - a_T)/4, & p_T &= (1 - a_C - a_G + a_T)/4. \end{aligned}$$

The stochastic space Θ_0 used here requires $0 < a_C, a_G, a_T < 1$ and all four transition entries positive. K2P is the submodel $a_C = a_T$, written $(1, s, g, s)$.

For a three-star tree with pendant vectors α, β, γ ,

$$q_{xyz}^T = \alpha_x \beta_y \gamma_z \quad (x + y + z = A).$$

At a network, each reticulation selects one parent and the distribution is the inheritance-weighted mixture of displayed trees.

The common rooted calculation topology has arcs

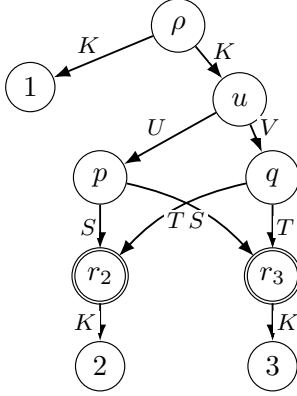
$$\rho \rightarrow 1, \rho \rightarrow u, u \rightarrow p, u \rightarrow q, p \rightarrow r_2, q \rightarrow r_2, p \rightarrow r_3, q \rightarrow r_3, r_2 \rightarrow 2, r_3 \rightarrow 3. \quad (1)$$

Both reticulations use weights $1/2, 1/2$. After suppressing the degree-two root, the maximal cyclic core consists of paths

$$p - u - q, \quad p - r_2 - q, \quad p - r_3 - q.$$

It has three incident leaf edges and two reticulations, hence is a strict level-two nontrivial 3-blob. Root suppression composes the two root-adjacent edge vectors coordinatewise.

(a) Rooted calculation network



both reticulations have inheritance weights $1/2, 1/2$

(b) Suppressed semi-directed trinet

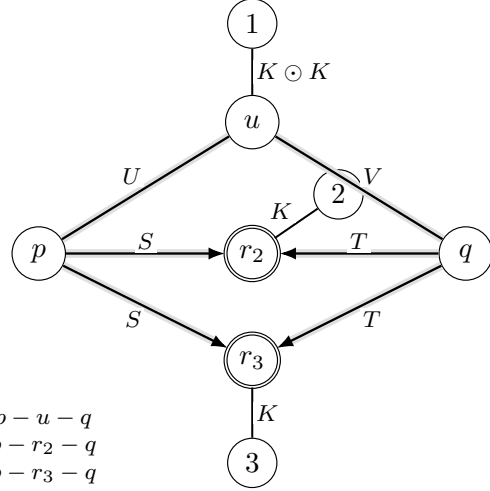


Figure 1: The rooted representation used for displayed-tree calculations and the suppressed semi-directed theta trinet. The labels K, U, V, S, T are instantiated separately in the K2P and K3P constructions.

Remark 1 (Two-terminal terminology). The collision theorem needs only the unambiguous fact that the whole theta core is a maximal nontrivial 3-blob. The source paper's printed boundary-vertex clauses for a 2-sub-blob and its degree-two suppression instruction create a wording tension on individual theta edges: each literal edge candidate contracts to four external incidences. Under the edge-incidence condition compatible with degree-two suppression, no proper induced theta subgraph qualifies. The topology verifier records both enumerations; neither interpretation removes the original theta distribution from the model.

2.1 Strict continuous-time cones

For K3P, symmetric continuous-time rates give

$$a_C = e^{-2(\lambda_G + \lambda_T)}, \quad a_G = e^{-2(\lambda_C + \lambda_T)}, \quad a_T = e^{-2(\lambda_C + \lambda_G)}.$$

All three rates are positive exactly when

$$a_C > a_G a_T, \quad a_G > a_C a_T, \quad a_T > a_C a_G. \quad (2)$$

For K2P, $a = (1, s, g, s)$, these reduce to

$$g > s^2. \quad (3)$$

3 An exact K2P collision over $\mathbb{Q}(\sqrt{71})$

Assign

$$\begin{aligned} K &= (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), & U &= (1, \tfrac{4}{5}, \tfrac{19}{30}, \tfrac{4}{5}), & V &= (1, \tfrac{7}{240}, \tfrac{239}{360}, \tfrac{7}{240}), \\ S &= (1, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), & T &= (1, \tfrac{1}{3}, \tfrac{1}{27}, \tfrac{1}{3}). \end{aligned} \quad (4)$$

Use K on $\rho \rightarrow 1, \rho \rightarrow u, r_2 \rightarrow 2, r_3 \rightarrow 3$; U, V on $u \rightarrow p, u \rightarrow q$; and S, T on arcs from p, q .

Lemma 2 (K2P edge admissibility). *Every rooted and effective network edge lies in Θ_0 , with strictly positive transition probabilities. The suppressed pendant edge is $K \odot K = (1, 1/4, 1/4, 1/4)$ with transition row $(7/16, 3/16, 3/16, 3/16)$.*

Proof. The transition rows of K, U, V, S, T are

$$(5/8, 1/8, 1/8, 1/8), (97/120, 11/120, 1/120, 11/120), (31/72, 121/1440, 289/720, 121/1440), \\ (1/2, 1/8, 1/4, 1/8), (23/54, 13/54, 5/54, 13/54).$$

All entries and nontrivial eigenvalues are strictly admissible. $\square$

For $x + y + z = A$, write $x = y + z$ and strip the four K factors. The mixture is

$$M_{y,z} = \frac{1}{4}(S_y S_z U_x + S_y T_z U_y V_z + T_y S_z U_z V_y + T_y T_z V_x). \quad (5)$$

Let $s = \sqrt{71}$ and set

$$P = (1, \frac{151}{36s}, \frac{107}{162}, \frac{151}{36s}), \quad R = (1, \frac{s}{40}, \frac{31}{120}, \frac{s}{40}). \quad (6)$$

Lemma 3 (Exact K2P core factorization). *For all $y, z \in G$, $M_{y,z} = P_{y+z} R_y R_z$.*

Proof. Direct substitution gives

$$M = \begin{pmatrix} 1 & 151/1440 & 3317/19440 & 151/1440 \\ 151/1440 & 71/1600 & 4681/172800 & 7597/259200 \\ 3317/19440 & 4681/172800 & 961/14400 & 4681/172800 \\ 151/1440 & 7597/259200 & 4681/172800 & 71/1600 \end{pmatrix},$$

and multiplying (6) gives the same matrix. The certificate lists every displayed contribution. $\square$

Define

$$\alpha = K^{\odot 2} \odot P = (1, \frac{151s}{10224}, \frac{107}{648}, \frac{151s}{10224}), \quad \beta = \gamma = K \odot R = (1, \frac{s}{80}, \frac{31}{240}, \frac{s}{80}). \quad (7)$$

The bound $8 < s < 9$ verifies their eigenvalue and transition inequalities.

Theorem 4 (Exact K2P tree–theta collision). *The K2P theta distribution in (4) equals the tree distribution in (7). Every edge lies in Θ_0 , every transition probability is strictly positive, and both inheritance probabilities are $1/2$. The exact minimum site-pattern probability is $1188799/79626240 > 0$.*

Proof. For a consistent label,

$$q_{xyz}^{N_\theta} = K_x^2 K_y K_z M_{y,z} = (K_x^2 P_x)(K_y R_y)(K_z R_z) = \alpha_x \beta_y \gamma_z = q_{xyz}^T.$$

The forty-eight inconsistent coordinates vanish on both models. Fourier inversion gives all pattern equalities, positivity, normalization, and the minimum. $\square$

Remark 5 (Comparison with the source statements). Lemma 5.6 states that the displayed invariant is “zero on $\mathcal{M}_1$ and strictly positive on $\mathcal{M}_2$,” and then concludes model disjointness. Corollary 5.8 asserts the corresponding JC/K2P disjointness when one network has a nontrivial m -blob, $m \geq 3$, and the other does not. Under the definitions and K2P convention stated in [1], Theorem 4 conflicts with the K2P conclusions of both statements already at three leaves.

4 A fixed-order issue in the K2P induction

The polynomial is

$$Q = q_{AGG}q_{GAG}q_{CCA}^2 - q_{AAA}q_{GGA}q_{TCG}^2. \quad (8)$$

The level-one 3-sunlet calculation remains valid when the reticulation-adjacent leaf occupies the designated final coordinate: independent substitution reproduces

$$Q = (\text{positive monomial}) \delta(1 - \delta)d_G e_G(1 - f_G)^2 > 0.$$

The arbitrary-level proof fixes the exposed leaf in that position and needs both displayed-child invariants in the same order. A child can retain a reticulation adjacent to a different leaf, whose level-one positivity is available only after another relabelling.

At the strict continuous-time collision of Section 5, expose r_3 . One child retains r_2 . Its exact values satisfy

$$Q_{(1,2,3)} = -1.919971072382827 \dots \times 10^{-9} < 0, \quad Q_{(1,3,2)} = 3.428488326525925 \dots \times 10^{-9} > 0.$$

Thus positivity after some child-specific relabelling does not imply positivity in the parent’s fixed order. This diagnoses the proof failure; Theorem 4 independently disproves the statement.

For completeness, an independent rational theta point makes $Q < 0$ under all six leaf permutations. Write each K2P edge as the pair (s, g) corresponding to $(1, s, g, s)$ and take

	e_1	U	V	A_2	B_2		A_3	B_3	D_2	D_3
s	69/100	53/100	23/50	9/25	19/50	s	9/20	3/25	17/100	49/100
g	4/5	23/100	3/20	89/100	17/50	g	21/100	39/50	19/100	27/100

$$\delta_2 = 3/5, \quad \delta_3 = 11/50.$$

Here e_1 is the effective leaf-1 pendant edge, A_i, B_i are the two incoming vectors at r_i , and D_i is the pendant vector below r_i . Every edge lies strictly in Θ_0 . The six values occur in three equal pairs:

[illegible]

Its minimum pattern probability is $2920987217429243/2000000000000000000 > 0$. This point is not needed for the tree collision; it independently shows that no global fixed sign for this single ordering of Q can be used on the theta topology.

5 A strict continuous-time K2P collision

Let ℓ be the unique root in $1.073231219980 < \ell < 1.073231219981$ of

$$634127002560\ell^3 - 2160769703472\ell^2 + 1746884136303\ell - 169873318739 = 0. \quad (9)$$

Define

$$\begin{aligned} v &= \frac{73394329}{14503216}\ell^2 - \frac{1453474193}{248626560}\ell + \frac{4133719}{3669120}, \\ x &= -\frac{366971645}{99450624}\ell^2 + \frac{4259402513}{340973568}\ell - \frac{42362455}{5031936}. \end{aligned}$$

Use

$$K = (1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}), U = (1, \frac{3}{4}, v, \frac{3}{4}), V = (1, \frac{1}{16}, x, \frac{1}{16}), S = (1, \frac{1}{5}, \frac{1}{2}, \frac{1}{5}), T = (1, \frac{7}{30}, \frac{1}{15}, \frac{7}{30}). \quad (10)$$

Let $t = \sqrt{1423}$ and define

$$P_C = P_T = \frac{79}{4t}, \quad R_C = R_T = \frac{t}{240}, \quad R_G = \frac{\ell}{4},$$

$$P_G = \frac{73394329}{4762633008}\ell^2 + \frac{368713223407}{81645137280}\ell - \frac{4985775401}{1204882560}.$$

With $P_A = R_A = 1$, the same factorization holds and tree edges are $K^{\odot 2} \odot P, K \odot R, K \odot R$.

Theorem 6 (Strict continuous-time K2P collision). *These parameters give an exact tree-theta collision. Every rooted, effective, and tree edge satisfies $g > s^2$, all transition entries are positive, the smallest continuous-time margin is $11/900$, and*

$$0.0149867914232177 < p_{\min} < 0.0149867914232311.$$

Proof. Exact arithmetic in the basis $1, \ell, \ell^2, t, t\ell, t\ell^2$ verifies all factorization and sign identities. Independently, ordinary-state Markov pruning on all four displayed trees agrees exactly with the comparison tree and Fourier inversion for all sixty-four patterns. $\square$

6 K2P local rank and exact collision families

The simultaneous exchange $C \leftrightarrow T$ gives ten orbits among the sixteen consistent K2P coordinates; after $q_{AAA} = 1$, the affine K2P space has dimension nine. The tree model has rank six.

Proposition 7 (Full K2P rank at both collisions). *At the simple collision, a selected 9×9 theta minor has determinant*

$$-\frac{7^2 11^2 19 107 151^2 15013}{2^{60} 3^{25} 5^{10}} \neq 0. \quad (11)$$

The same minor at the strict continuous-time witness lies rigorously between $-4.129735 \times 10^{-22}$ and $-4.129729 \times 10^{-22}$.

The theta map is therefore dominant. The semi-directed theta has twenty parameters, while the tree germ has codimension three in the ambient space.

Corollary 8 (K2P local collision geometry). *Near the simple witness, theta parameters whose output lies in the tree model form a smooth 17-dimensional submanifold of codimension three. Generic theta parameters are not tree-equivalent, but full pointwise distinguishability fails.*

There is also an exact family. Write

$$U = (1, u, v, u), \quad V = (1, w, x, w), \quad S = (1, a, b, a), \quad T = (1, c, d, c).$$

The relevant entries are

$$\begin{aligned} M_{AC} &= (au + cw)/2, & M_{AG} &= (bv + dx)/2, \\ M_{CC} &= (a^2 + 2acuw + c^2)/4, & M_{GG} &= (b^2 + 2bdvx + d^2)/4, \\ M_{CG} &= (abu + adux + bcvw + cdw)/4, & M_{CT} &= (a^2v + 2acuw + c^2x)/4. \end{aligned}$$

In the positive chamber, tree factorization is equivalent to

$$M_{CG}^2 - M_{AC}^2 M_{GG} = 0, \quad M_{CT}^2 M_{GG} - M_{AG}^2 M_{CC}^2 = 0. \quad (12)$$

Indeed, positivity selects the unique recovery

$$R_C = \sqrt{M_{CC}}, \quad R_G = \sqrt{M_{GG}}, \quad P_C = M_{AC}/R_C, \quad P_G = M_{AG}/R_G;$$

the two equations in (12) are exactly the remaining CG and CT factorization conditions, with no sign ambiguity. At the simple witness the Jacobian with respect to (v, x) is

$$675554683609333/194995116803358720000000 > 0.$$

Corollary 9 (Exact symmetric collision family). *The simple K2P witness lies on a smooth local six-dimensional algebraic family of positive core parameters satisfying (12).*

7 An exact K3P collision

Let $h = 5^{-1/4}$. On the same topology assign

$$\begin{aligned} K &= (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), & U &= (1, \tfrac{h}{3}, h, \tfrac{1}{3}), & V &= (1, h, \tfrac{h}{3}, \tfrac{1}{3}), \\ S &= (1, \tfrac{3h^2}{4}, \tfrac{1}{4}, \tfrac{3}{10}), & T &= (1, \tfrac{1}{4}, \tfrac{3h^2}{4}, \tfrac{3}{10}). \end{aligned} \quad (13)$$

All inverse-Fourier rows are strictly positive. Define

$$B = (1, \tfrac{h^2}{2}, \tfrac{h^2}{2}, \tfrac{h^2}{2}), \quad P = (1, \tfrac{5h^3+h}{4}, \tfrac{5h^3+h}{4}, h^2). \quad (14)$$

Exact reduction modulo $5h^4 - 1$ gives $M_{y,z} = P_{y+z} B_y B_z$. The comparison tree has

$$\alpha = (1, \tfrac{5h^3+h}{16}, \tfrac{5h^3+h}{16}, \tfrac{h^2}{4}), \quad \beta = \gamma = (1, \tfrac{h^2}{4}, \tfrac{h^2}{4}, \tfrac{h^2}{4}). \quad (15)$$

Theorem 10 (Exact K3P tree-theta collision). *The theta distribution defined by (13) equals the tree distribution in (15). Every network, effective, and tree edge lies in Θ_0 with strictly positive transition entries. This gives a negative answer to the high-level K3P trinet question raised in [1].*

Proof. The factorization (14) gives $q_{xyz}^{N_\theta} = K_x^2 K_y K_z M_{y,z} = \alpha_x \beta_y \gamma_z$ on the sixteen consistent labels; all others vanish. Exact inversion gives all pattern equalities. $\square$

The quartic point follows from the positive ansatz

$$U = (1, b\tau, b, \tau), \quad V = (1, b, b\tau, \tau), \quad S = (1, \kappa d, d, e), \quad T = (1, d, \kappa d, e).$$

On the slice $\tau = 1/3$, the factorization equations force

$$\kappa = 3/\sqrt{5}, \quad b^2 = 1/\sqrt{5}, \quad e/d = 6/5,$$

so the positive branch has $b = 5^{-1/4}$.

8 K3P local overlap

The affine consistent K3P space with $q_{AAA} = 1$ has dimension fifteen.

Proposition 11 (Full-rank K3P local overlap). *At the exact collision,*

$$\boxed{\det J_* = \frac{h(10h^2 + 1)}{2^{61}3^45^{14}} > 0.} \quad (16)$$

Hence the fixed theta image contains an ordinary open neighborhood in the full fifteen-dimensional affine space. The tree model has rank nine. In the twenty-nine-dimensional semi-directed theta parameter space, the local collision locus is smooth of dimension 23 and codimension six.

Proof. Exact differentiation and Bareiss elimination give (16); $2/3 < h < 7/10$ proves its sign. The tree rank follows from three independent positive 3×3 blocks. The preimage theorem gives $29 - (15 - 9) = 23$. $\square$

This does not mean generic theta parameters map to trees: the collision locus has codimension six.

9 Strict continuous-time K3P extension

The closed-form witness satisfies two boundary equalities,

$$U_C = U_G U_T, \quad V_G = V_C V_T,$$

with all other inequalities in (2) strict. Increase U_C and V_G by a common $\varepsilon > 0$ and solve for fifteen pivot parameters while fixing the fifteen nonconstant outputs. Proposition 11 supplies the invertible derivative required by the real-analytic implicit-function theorem.

The exact tangent certificate verifies

$$J_* p'(0) + F_{U_C} + F_{V_G} = 0, \quad \frac{da_G^U}{d\varepsilon} = \frac{6}{1 + 10h^2}.$$

The two saturated margins have derivatives

$$\frac{10h^2 - 1}{1 + 10h^2} > 0, \quad 1 > 0.$$

All other inequalities persist by continuity.

Corollary 12 (Strict continuous-time K3P collision). *The same fixed tree distribution has a nearby preimage on the theta topology for which every network edge has all three instantaneous K3P substitution rates strictly positive.*

The exact certificate verifies the analytic theorem's algebraic hypotheses and tangent data; the existence of the branch is the conclusion of the implicit-function theorem.

10 Scope, implications, and open questions

The level-one K2P 3-sunlet calculation in Lemma 4.1 of [1] is not contradicted by these examples. The new conclusion is narrower: universal three-leaf tree–trinet distinguishability fails on the strict level-two theta topology. The full-rank calculation shows that the fixed theta map is dominant onto the nine-dimensional affine K2P space, while the tree variety has dimension six. Hence tree equivalence is nongeneric among theta parameters, yet the positive collision set is locally 17-dimensional rather than isolated or confined to the stochastic boundary.

For K3P, the level-one result is likewise not addressed, while strict level two admits exact and locally full-dimensional overlap in distribution space. Thus under both Kimura models a nontrivial three-way blob is not universally detectable from a three-leaf distribution. No JC collision or analogous proof failure is established here.

The work does not decide whether a useful replacement statement holds with four selected leaves, restricted blob topology, generic rather than pointwise parameters, a vector of several leaf-order invariants, or an equivalence relation generated by local theta moves.

11 Future classification problems

1. Classify the complete K2P and K3P tree–theta intersections and positive chambers.
2. Determine whether the two mechanisms belong to one symmetry-breaking family.
3. Identify local observational-equivalence moves and whether they compose independently across blobs.
4. Decide K2P and K3P quartet distinguishability and corrected level-two identifiability.
5. Determine whether JC genuinely excludes theta/tree collisions or requires another topology.
6. Characterize continuous-time, clock, or edge-sharing restrictions that restore full distinguishability.

12 Exact verification and provenance

The short command `python3 verify_k2p_simple.py` checks the compact witness with the standard library. The complete command `python3 verify.py` reconstructs source conventions, both K2P collisions, the induction-order audit, direct Markov pruning, both K2P rank calculations, the exact family, the K3P collision, rank-15 determinant, and analytic-IFT tangent data. It ends with `ALL EXACT CHECKS PASSED`.

No public posting or submission has occurred. AI-assisted analysis first identified a possible sign-definiteness and leaf-order issue in the K2P invariant and produced a high-precision collision candidate. The exact $\mathbb{Q}(\sqrt{71})$ witness, the strict continuous-time algebraic witness, the rank calculations, and the collision-family results were then constructed and independently verified. Exact certificates and replayable verifiers are supplied so the mathematical claims do not depend on accepting the discovery process.

References

- [1] J. Brits, N. Holtgreffe, L. van Iersel, and S. Martin, *On Tree–Network Distinguishability and Full Identifiability of Phylogenetic Networks*, arXiv:2607.12919v2 [q-bio.PE], revised 29 July 2026; manuscript dated 30 July 2026.